BIRZEIT UNIVERSITY

Electrical and Computer Engineering Department
ENEE2313, Signals and Systems, Assignment

Matlab Assignment solution

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Section: 3.

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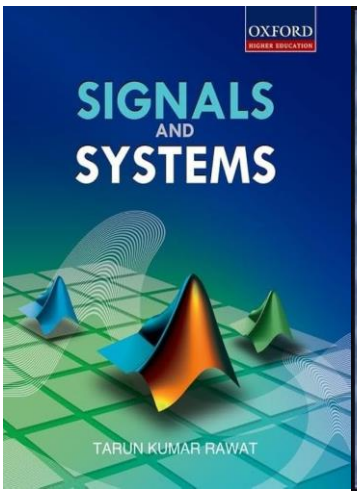
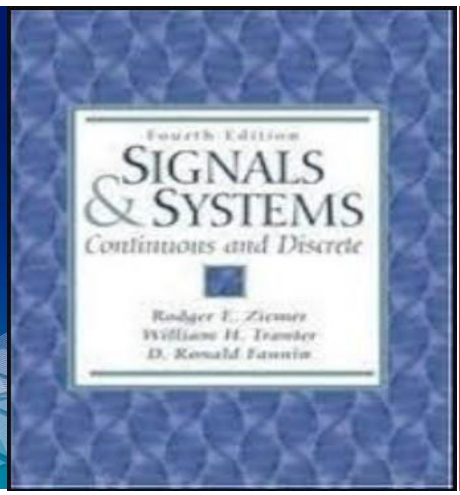
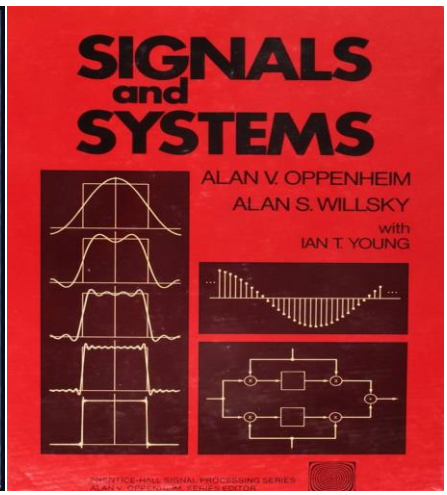






Problem 1:

A periodic signal is defined by the following:

$$x(t) = 10\sin(\omega_0 t) \sum_{n=-\infty}^{\infty} \left[\Pi\left(\frac{t - \frac{3\,T_0}{4} - nT_0}{\frac{T_0}{4}} \right) + \Pi\left(\frac{t + \frac{3\,T_0}{4} + nT_0}{\frac{T_0}{4}} \right) \right]$$

The first two periods: n = 0 and n = 1

w0 = (2 * pi) / To

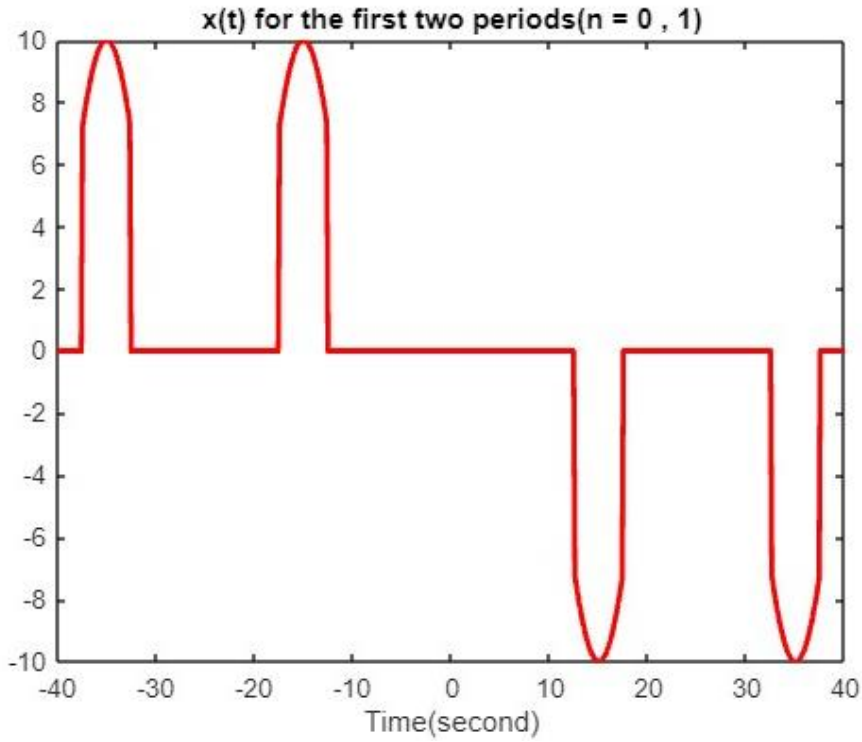
1- Using input command to enter the value of T0, and plot the first two periods of the signal.

- **Code:**

```
T = input('enter the value of T0 :' ) ;  %input command to enter the value of T0
w = (2 * pi) / T ;

TimeInterval = (-2*T):0.01:(2*T) ;

%Calculate shifted times for n = 0 and n = 1:
t1_n0 = (TimeInterval - ((3 * T)/4)) / (T/4) ;
t2_n0 = (TimeInterval + ((3 * T)/4)) / (T/4) ;

t1_n1 = (TimeInterval - ((3 * T)/4) - T) / (T/4) ;
t2_n1 = (TimeInterval + ((3 * T)/4) + T) / (T/4) ;

%The periodic signal:
x = 10 * sin(w * TimeInterval) .* (rectpuls(t1_n0) +
rectpuls(t2_n0) + rectpuls(t1_n1) + rectpuls(t2_n1));

plot(TimeInterval , x , 'r' , 'LineWidth' , 2) ;
```

```
title('x(t) for the first two periods(n = 0 , 1)') ;
xlabel('Time(second)') ;
```

- **Figure:**
  When T0 = 20



2- Obtain Fourier Series of the Signal, and plot the first two periods of the signal.

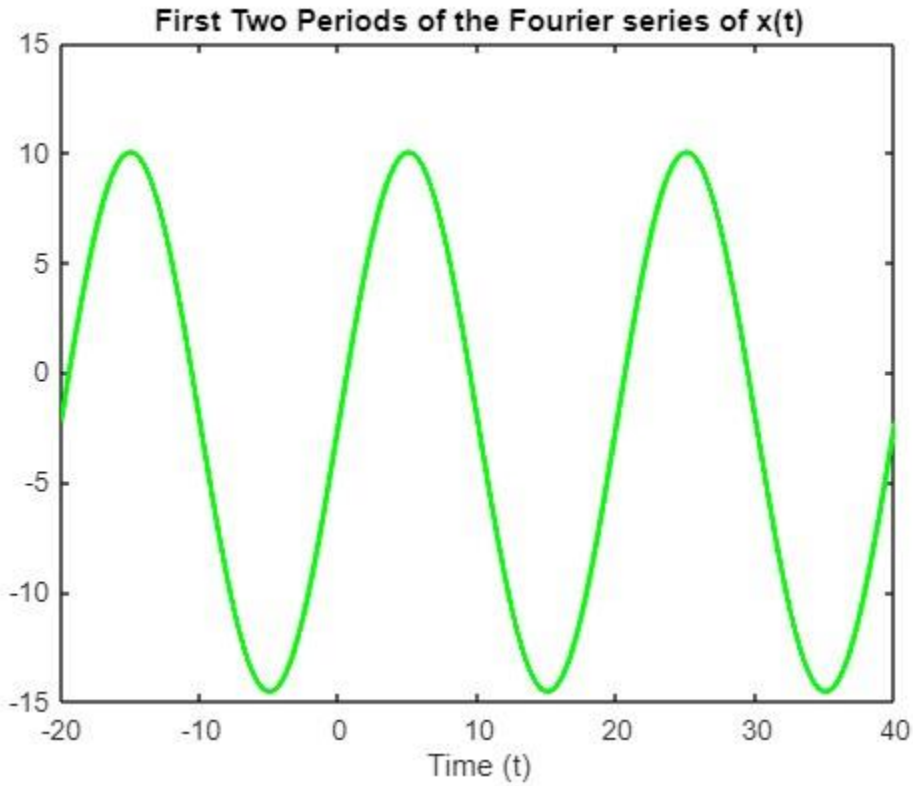
- **Code:**

```
T0 = input('Enter the value of T0: ') ;  %input command to enter the value of T0
w0 = (2 * pi) / T0 ;

TimeInterval = (-T0):0.01:(2 * T0) ;

%Calculate shifted times for n = 0 and n = 1:
t1_n0 = (TimeInterval - ((3 * T0)/4)) / (T0/4) ;
t2_n0 = (TimeInterval + ((3 * T0)/4)) / (T0/4) ;
```

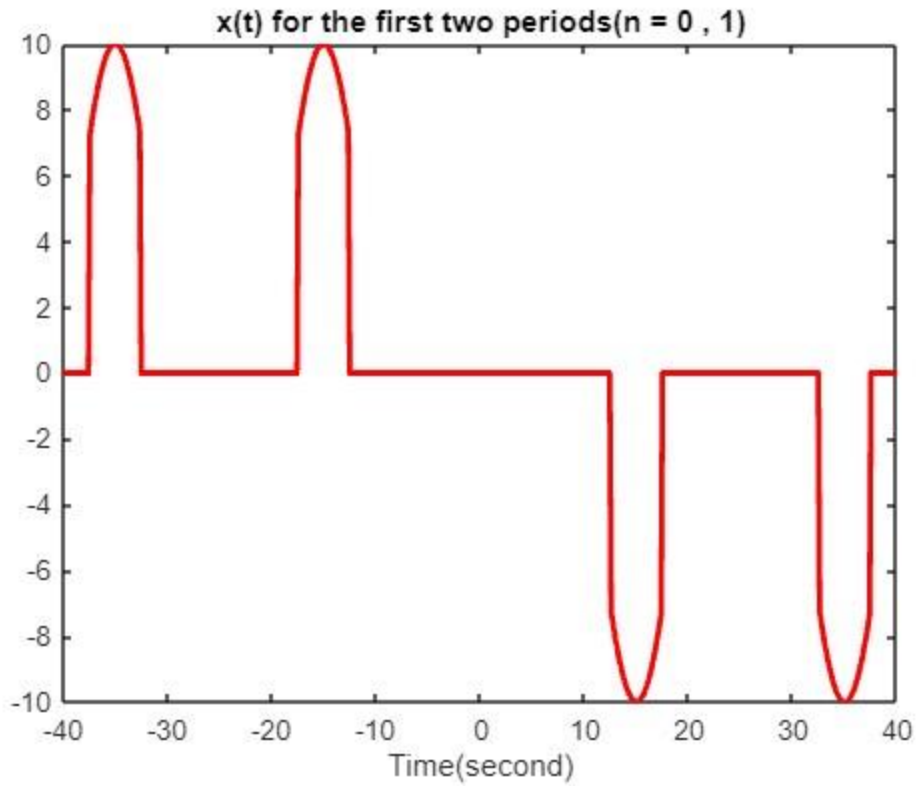
```
t1_n1 = (TimeInterval - ((3 * T0)/4) - T0) / (T0/4) ;
t2_n1 = (TimeInterval + ((3 * T0)/4) + T0) / (T0/4) ;

%the periodic signal x(t):
x = 10 * sin(w0 * TimeInterval) .* (rectpuls(t1_n0) +
rectpuls(t2_n0) + rectpuls(t1_n1) + rectpuls(t2_n1)) ;

%Compute Fourier Series Coefficients:
N = 10 ;  %Number of Fourier terms you want to calculate
a0 = 1/T0 * trapz(TimeInterval , x) ;  %DC component

an = zeros(1 , N) ;
bn = zeros(1 , N) ;

for n = 1:N
   an(n) = 2/T0 * trapz(TimeInterval , x .*
cos(n*w0*TimeInterval));
   bn(n) = 2/T0 * trapz(TimeInterval , x .*
sin(n*w0*TimeInterval));
end

x = a0 + ((an(1) * cos(w0 * TimeInterval)) + (bn(1) * sin(w0 *
TimeInterval))) ;

plot(TimeInterval , x , 'g' , 'LineWidth' , 2) ;
title('First Two Periods of the Fourier series of x(t)') ;
xlabel('Time (t)') ;

%trapz is used for numerical integration using the trapezoidal rule.
```

%In this code, it's applied to approximate the integral needed to calculate the Fourier coefficients an and bn, where the integral is approximated as the area under the curve.

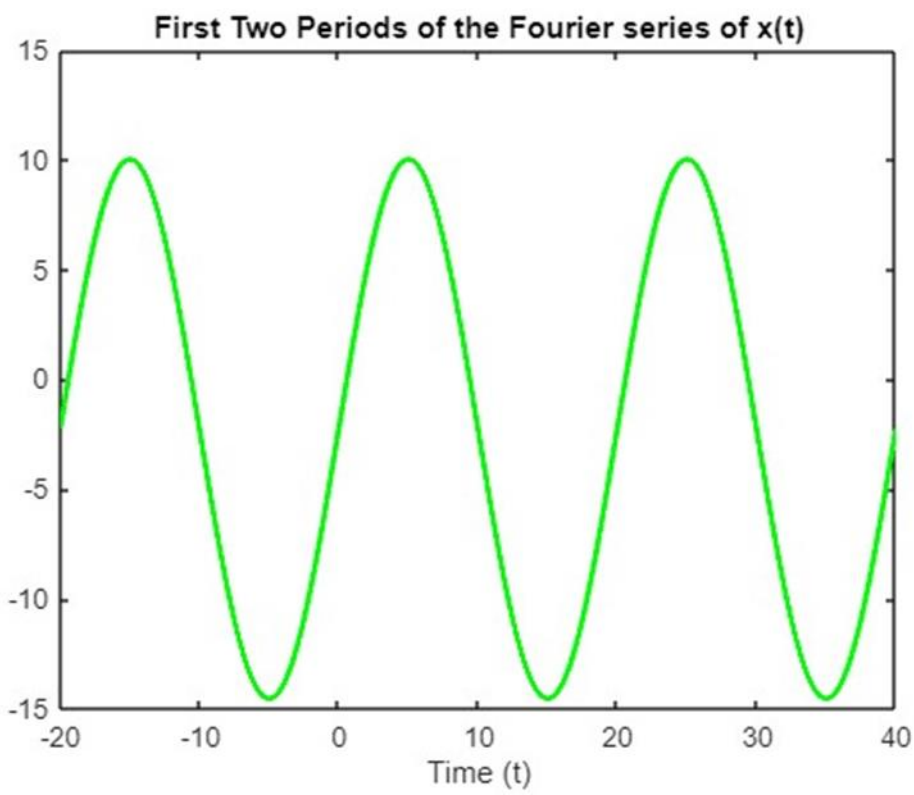
- **Figure:**
  When T0 = 20



3- Compare your results



x(t) is a sharp angular wave periodic signal



This is a smooth sinusoidal wave and a way to represent a complex periodic signal as a sum of simpler well-understood sinusoidal functions.

The Fourier series decomposes the original periodic signal into it's constituent frequencies and amplitudes.

The amplitudes of each harmonic depends on the shape of the original signal.

It consists of a fundamental frequency and it's harmonics.

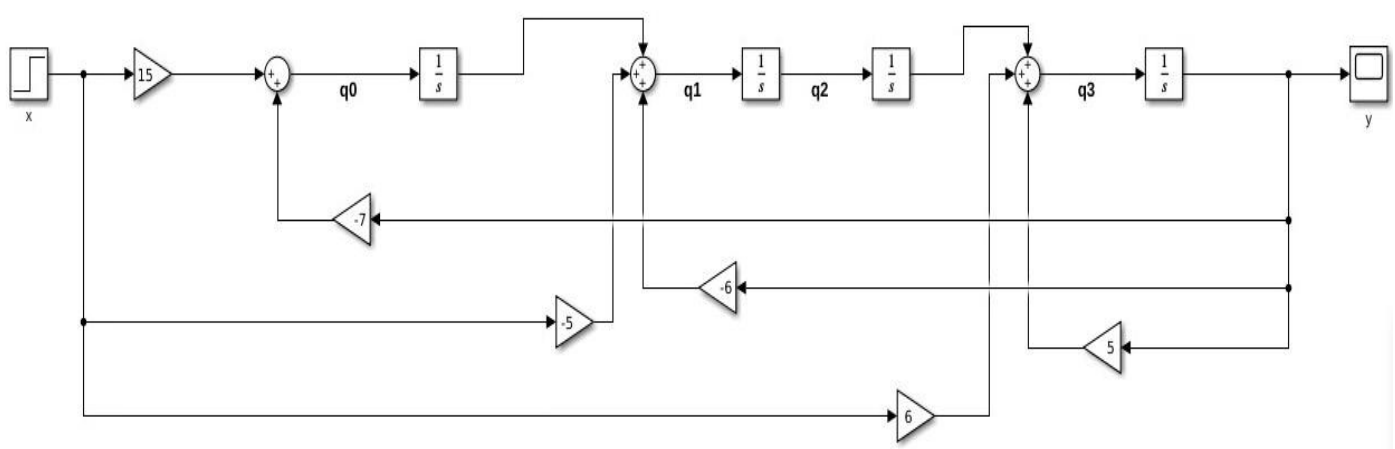
It's an approximation of the original signal and the accuracy improves as more harmonics are included.

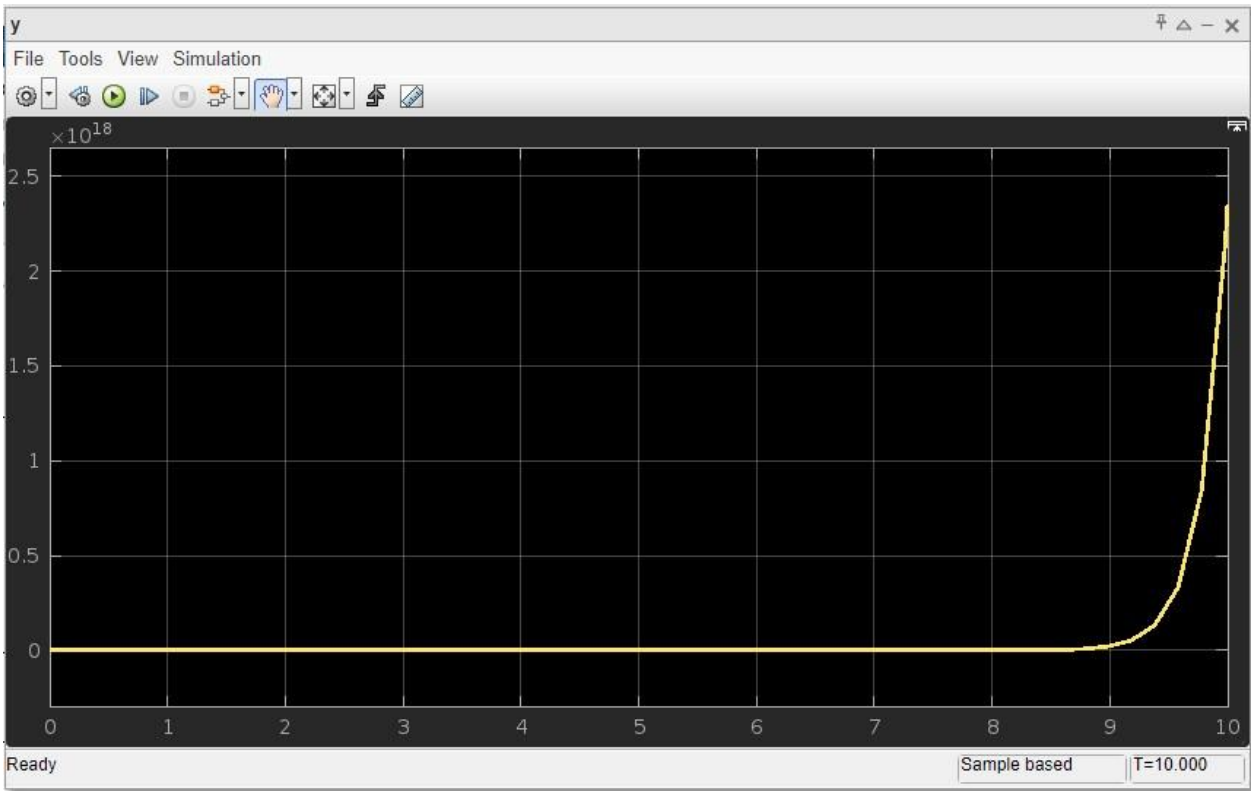
<u>Problem 2:</u>

Use Simulink (Matlab) to simulate the following system then show and plot the step response of the system.

$$\frac{d^4y(t)}{dt^4} - 5\frac{d^3y(t)}{dt^3} + 6\frac{dy(t)}{dt} + 7y(t) = 6\frac{d^3x(t)}{dt^3} - 5\frac{dx(t)}{dt} + 15\,x(t)$$

y’’’’ – 5y’’’ + 6y’ + 7y = 6 x’’’ – 5x’ + 15x
y’’’’ – 5y’’’ + 6y’ – 6x’’’ + 5x’ = 15x – 7y = q0
y’’’ – 5y’’ + 6y – 6x’’ + 5x = integration(q0)
y’’’ – 5y’’ – 6x’’ = integration(q0) – 6y – 5x = q1
y’’ – 5y’ – 6x’ = integration(q1) = q2
y’ – 5y – 6x = integration(q2)
y’ = integration(q2) + 5y + 6x = q3
y = integration(q3)




y
File Tools View Simulation
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Sample based
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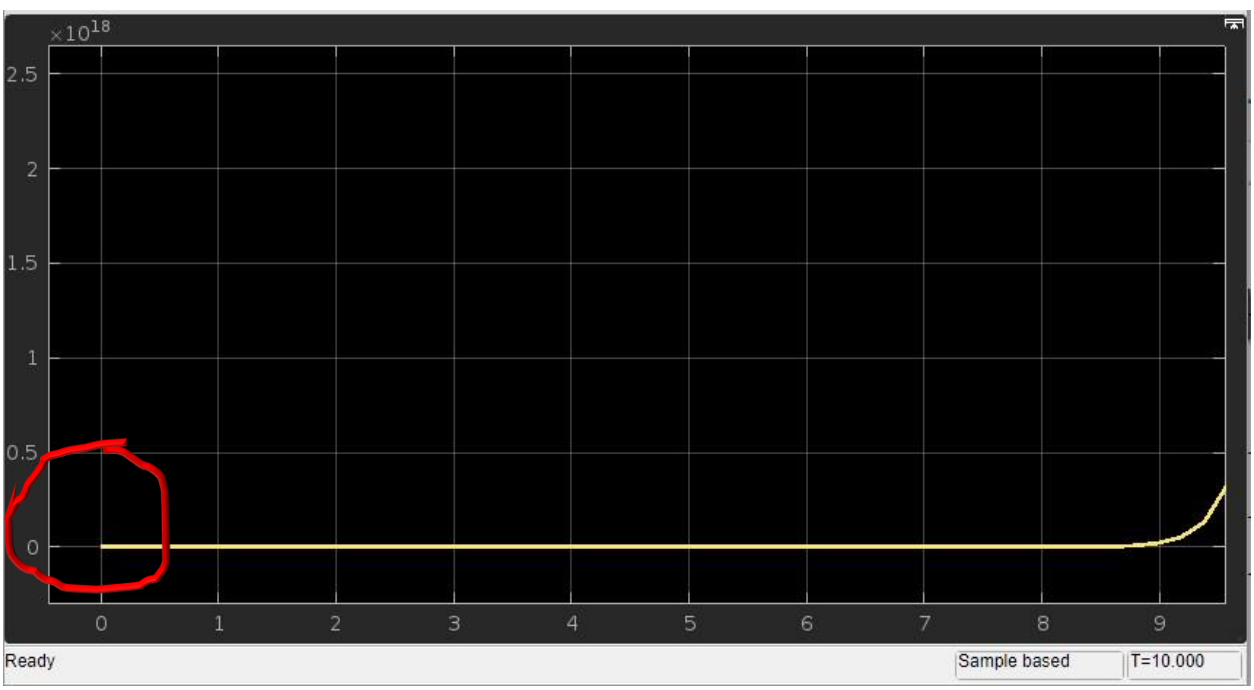
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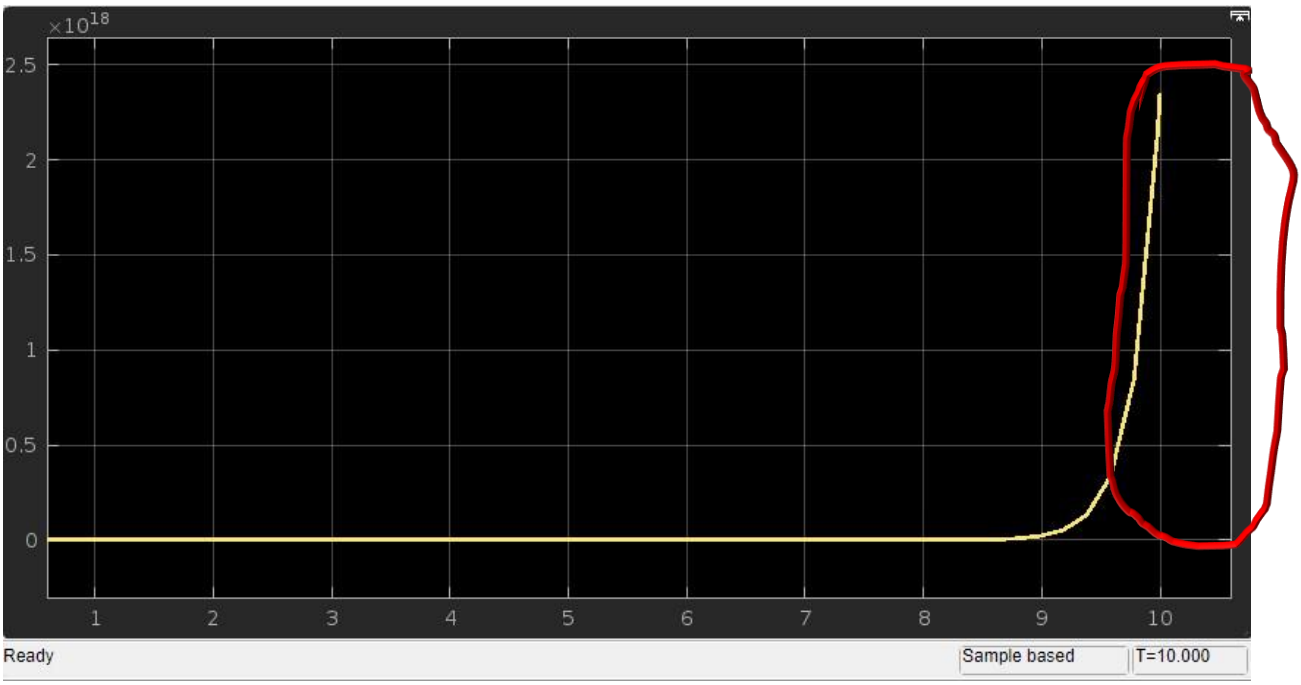
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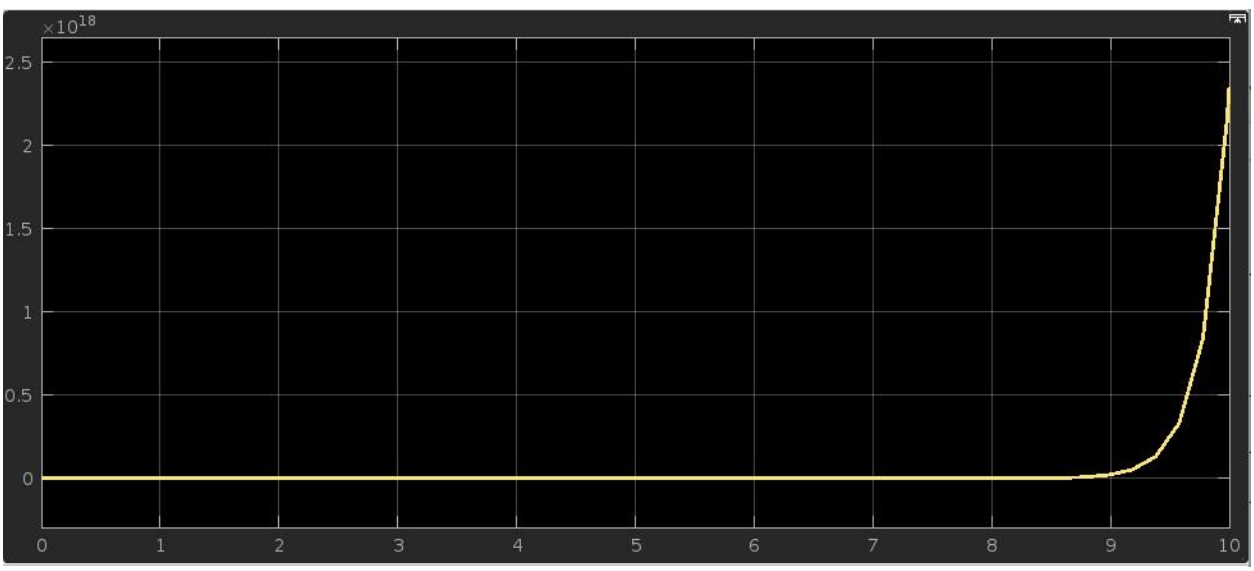
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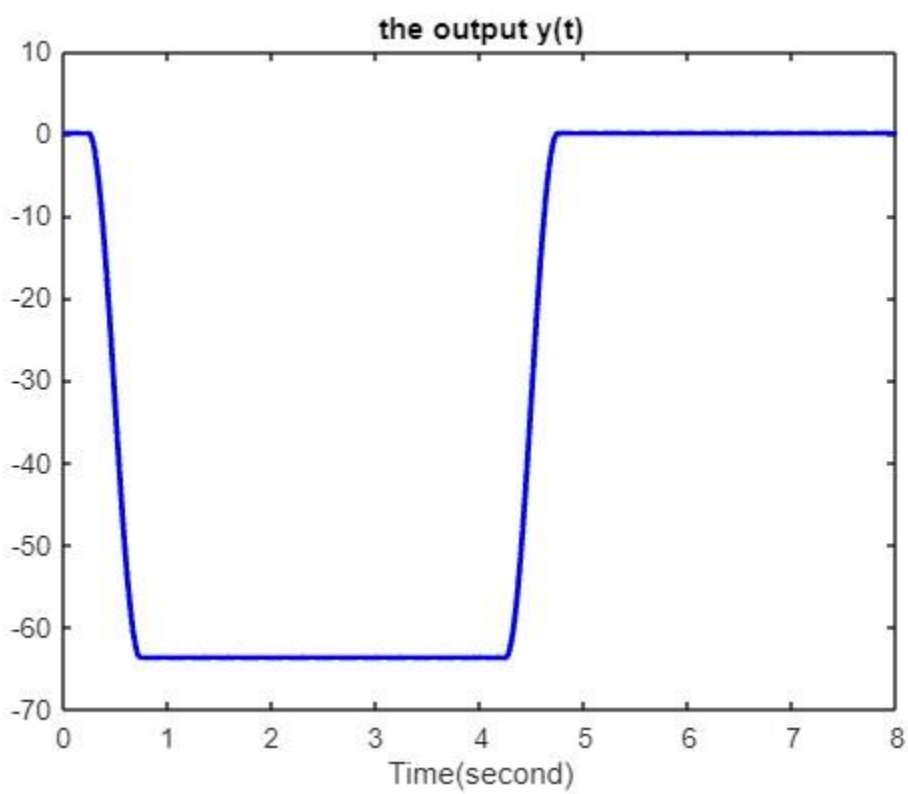
Problem 3:

Write a program that computes and plots the output of the LTI Systems functions with impulse response *h*(t) = 2Π ((t – 2) /4) to the input x(t) = cos(2*pi*t) Π ((t – 0.5) /0.5).

y(t) = x(t) * h(t): convolution

- **Code:**

```
TimeInterval = 0:0.01:4 ;
x = cos(2 * pi * TimeInterval) .* rectpuls((TimeInterval - 0.5)/0.5);
h = 2 * rectpuls((TimeInterval - 2)/4) ;
y = conv(x , h) ;
TimeInterval = 0:0.01:8 ;
plot(TimeInterval , y , 'b' ' , 'LineWidth' , 2) ;
title('the output y(t)') ;
xlabel('Time(second)') ;
```

- **Figure:**